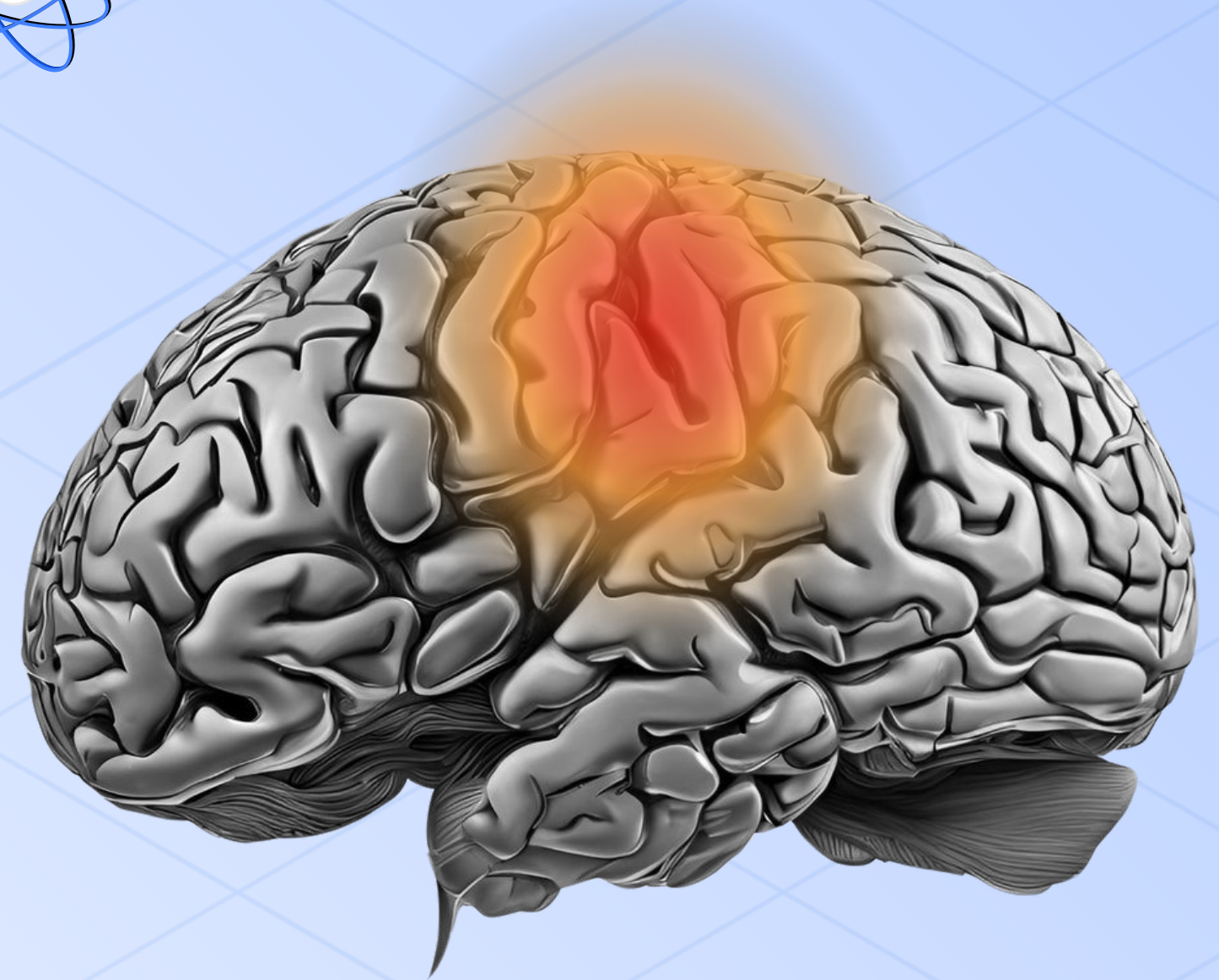
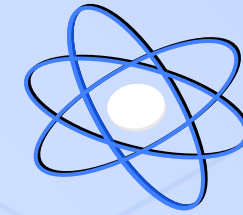


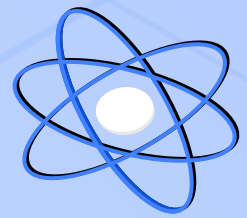
Decoding Motor Intent from Sensorimotor Oscillations



Keerat Kaur, Nicolas Thiele, Alex Jacobo

Professor Mukamel, COGS 118C: WI 2026

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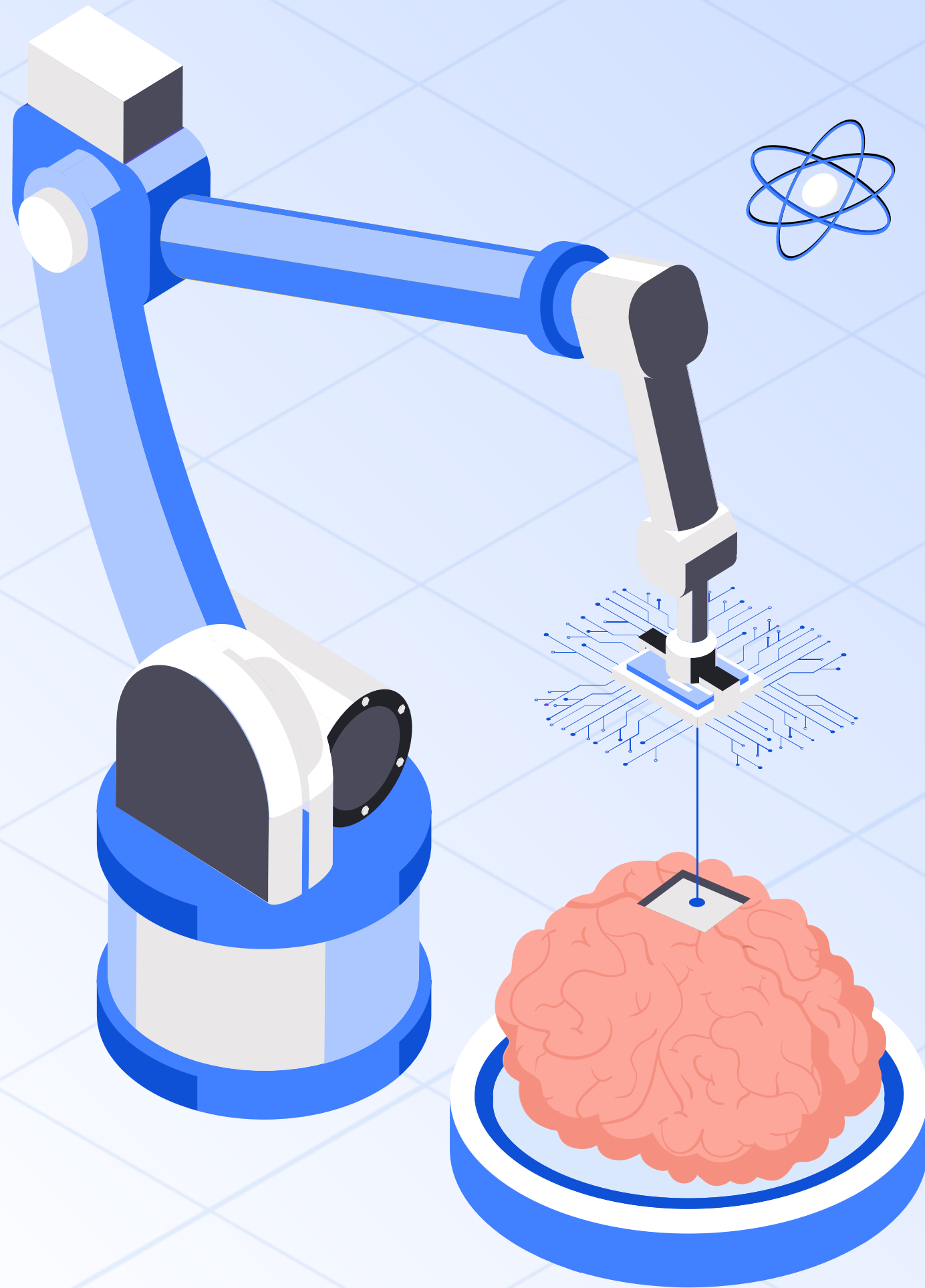
Results

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Conclusion

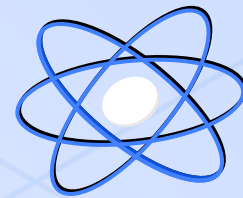


Introduction



Motivations & *Goals*

Why?

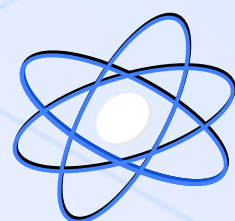
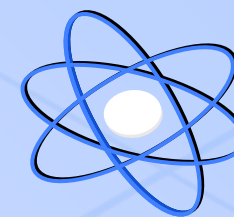
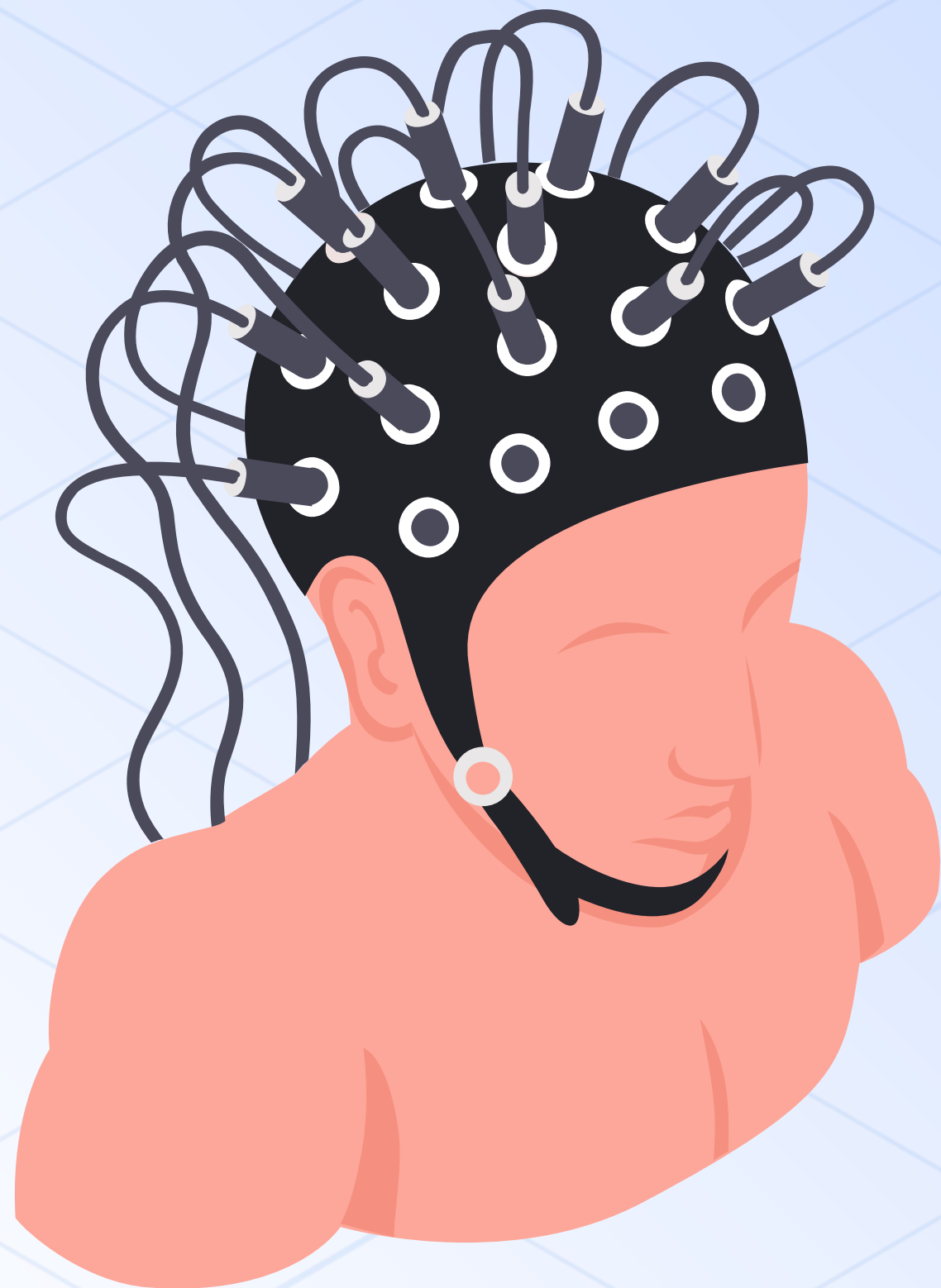


- Detecting Motor Intent for BCI development
- EEG Pros
- Well-established markers for ERD in Mu and Beta bands

Objectives

- Distinguish reliable differences between rest and motion intent using spectral features
- Measure ERD in the mu/beta band over motor cortex (C3, Cz, C4)
- Compare Intent with Baseline and PSD and Spectral Analysis

Data



PhysioNet EEG Motor Movement/Imagery Dataset

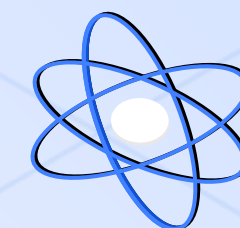
- Original 109 subjects
- Used Motor Movement (+) Motor Imaging tasks
- 64 channel EEG recordings
- 160 HZ sampling rate
- Baseline Rest
- L/R Hand movement
- L/R first movement

Utilizations:

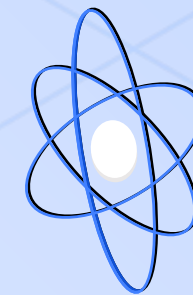
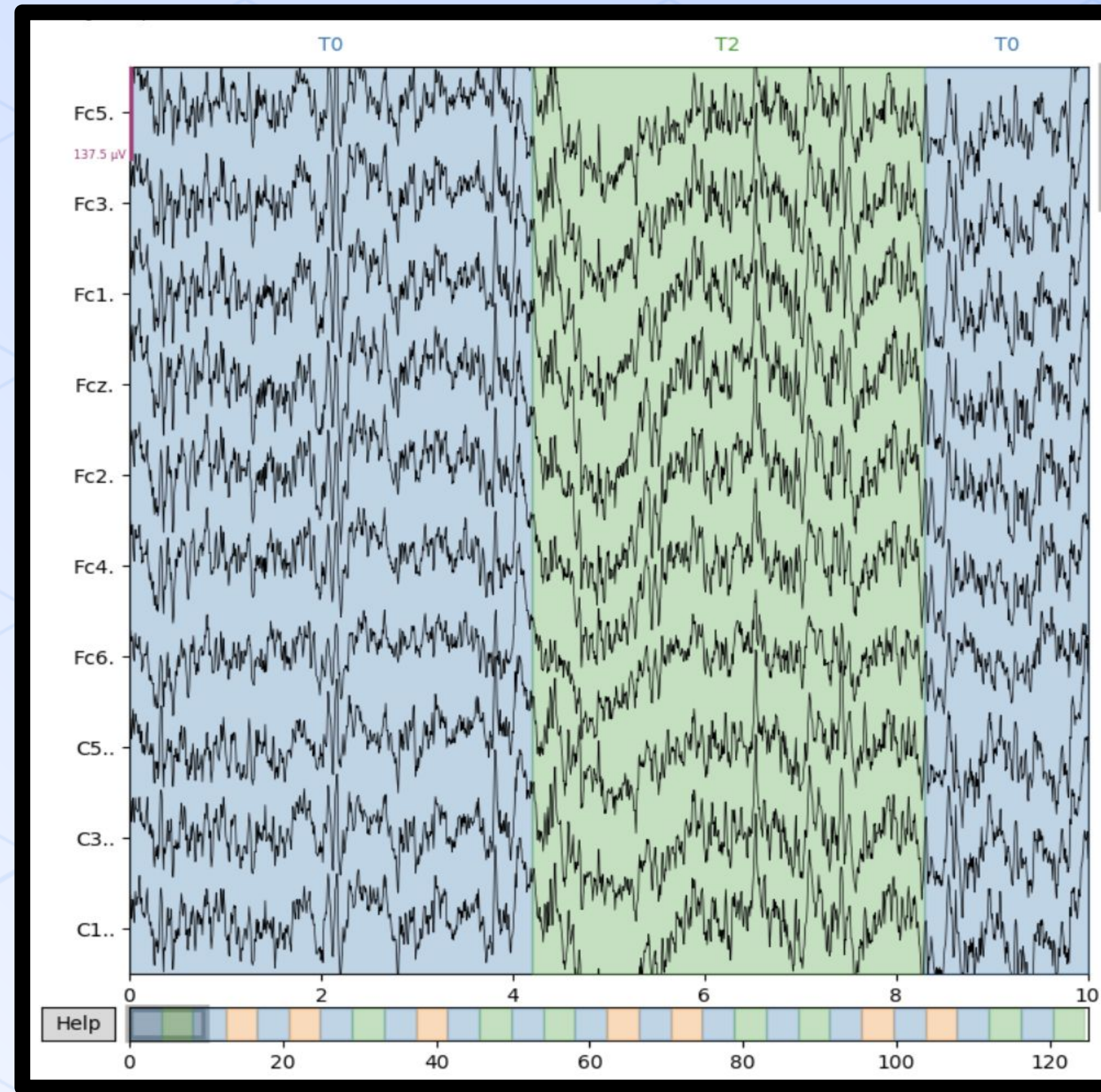
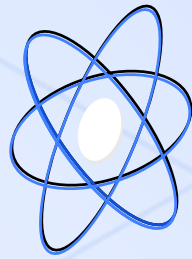
- Focus : 6 subjects
- 6 imagery runs

Event Structures:

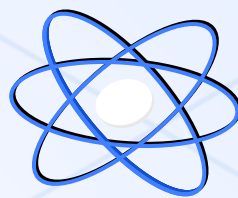
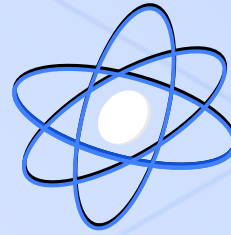
- T0 = Rest
- T1= L Hand Imagery
- T2= Right Hand imagery

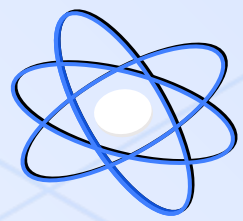


Raw EEG signal



Signal Processing *Methods*





Preprocessing / Analysis

EEG Preprocessing

- Channel selection (C3, Cz, C4)
- Bandpass filtering (1–40 Hz)
- Trial segmentation around movement onset

Spectral Analysis

Power spectral density (PSD) is estimated using Welch's method to compare oscillatory power across conditions.

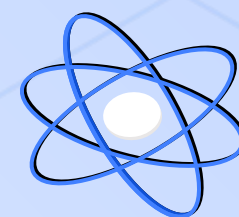
Time–Frequency Analysis to visualize task-related power changes over time

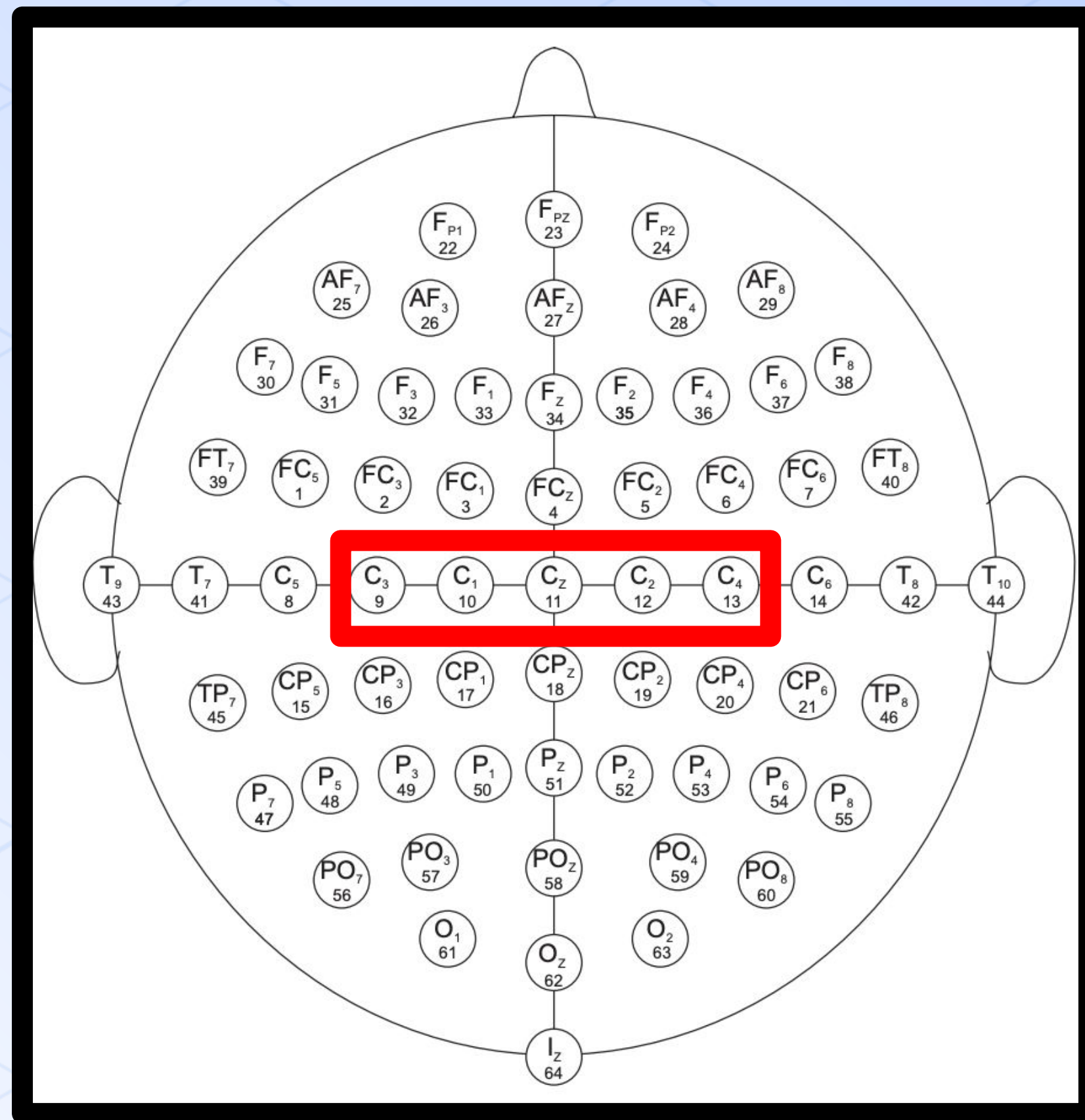
Morlet transforms are used to visualize changes in oscillatory power across time.
Event-Related Desynchronization (ERD)

ERD is calculated by comparing baseline power to task power:

$$\text{ERD (\%)} = ((\text{Baseline Power} - \text{Task Power}) / \text{Baseline Power}) \times 100$$

This quantifies the suppression of mu and beta rhythms during movement.



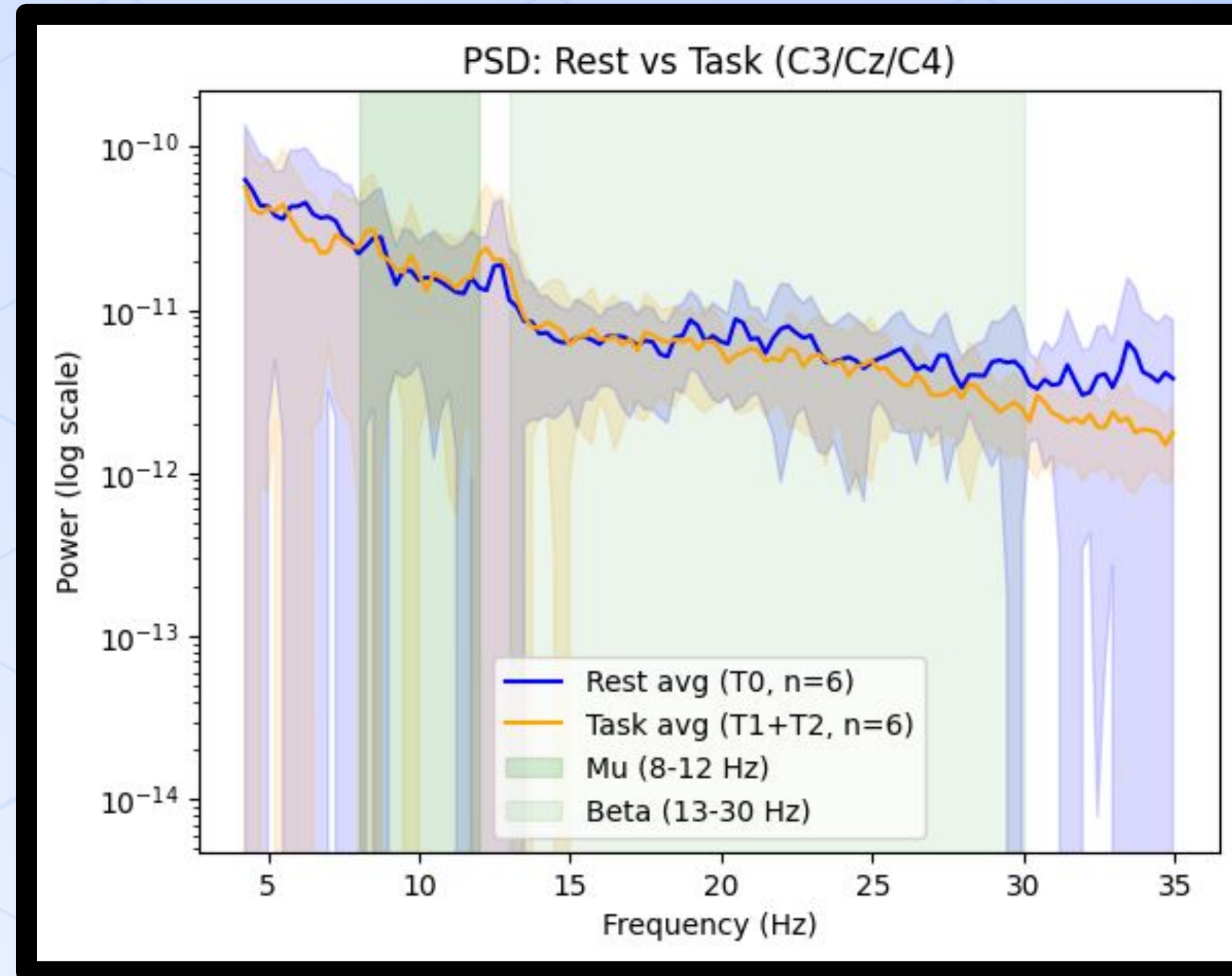


Results

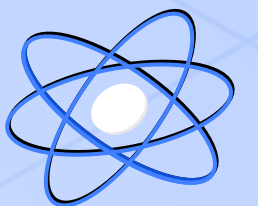


PSD Rest vs Task

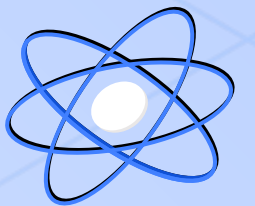
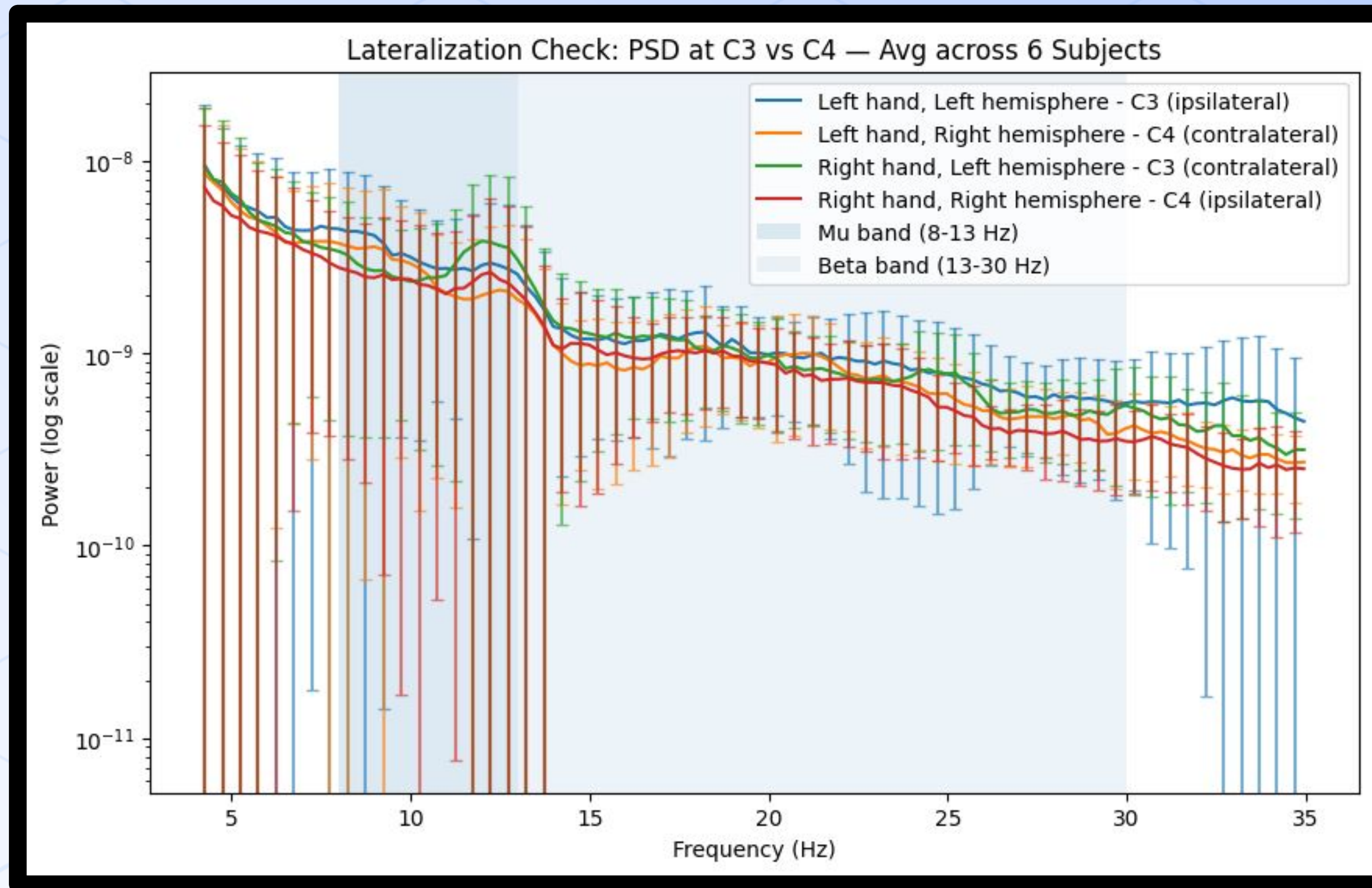
- Shaded area = (SD) standard deviation across all 6 subjects
- . Mu band: $p = 0.089$, trending, not significant
- . Beta band: $p = 0.089$ across 6 subjects, trending, not significant



- noted in previously established literature, in a phenomenon called BCI inefficiency, subjects cannot produce reliable mu scores unlike beta where they are highly concentrated around each other.

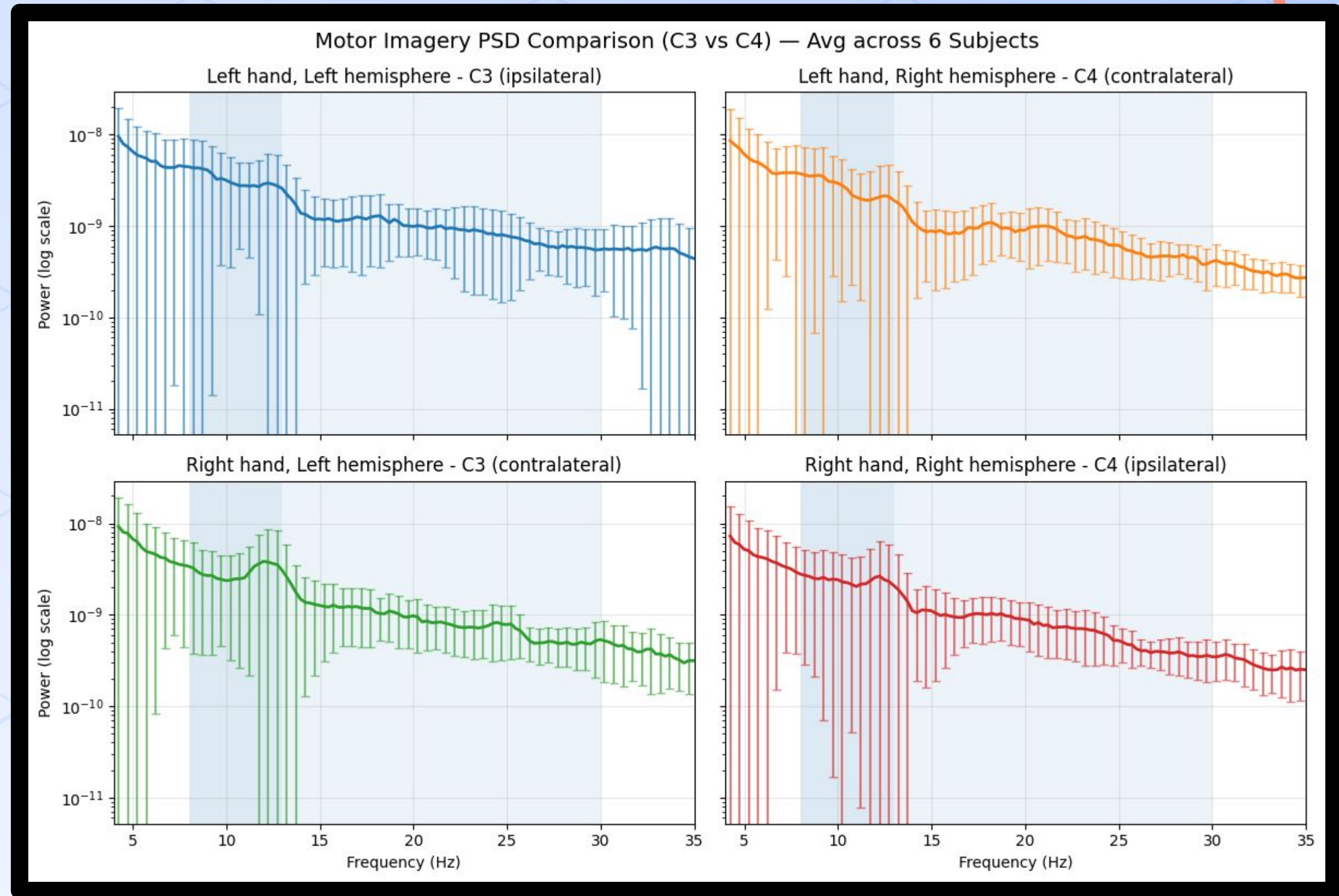


Lateralization



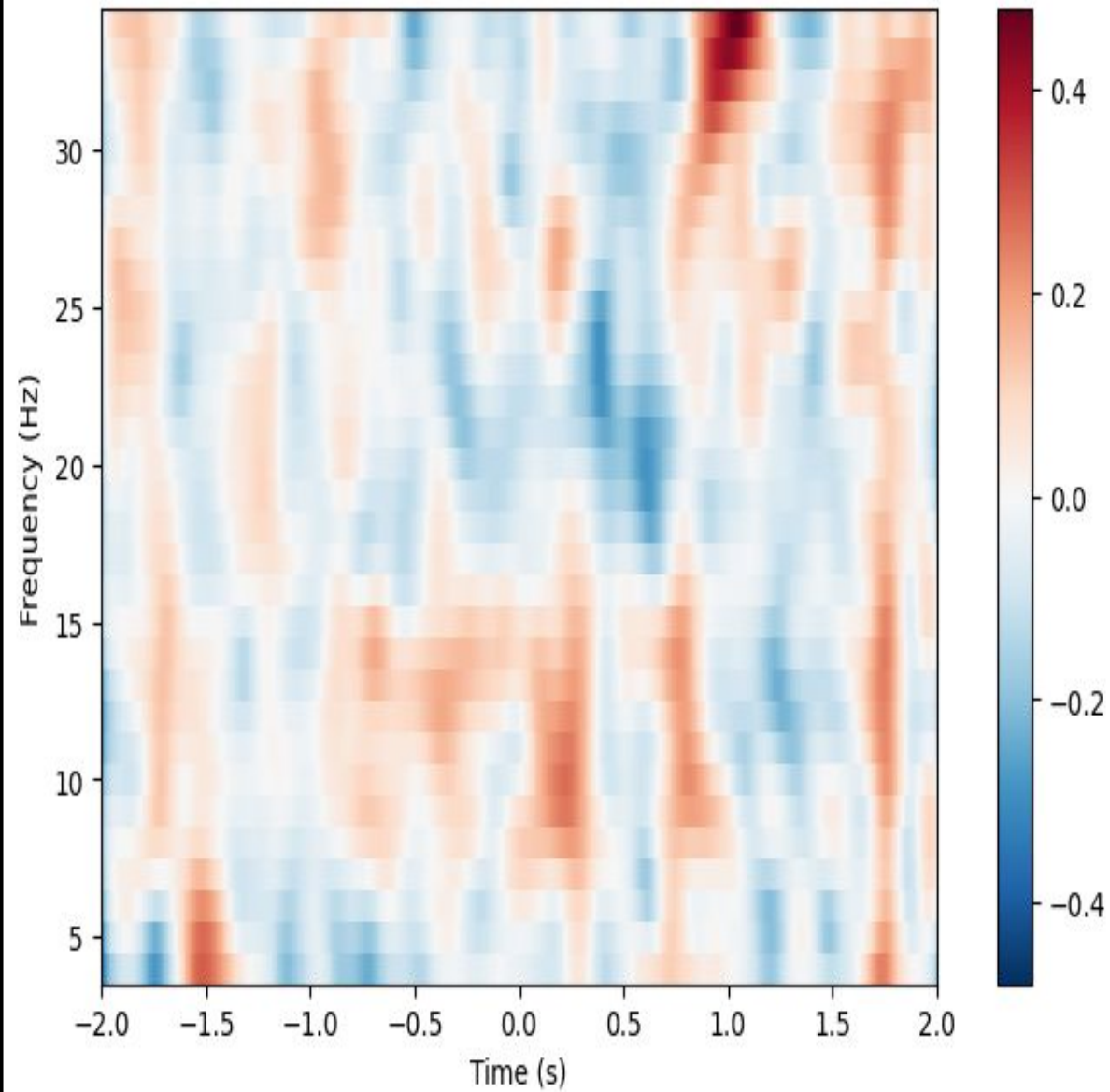
Lateralization

- Error bars indicated the (SD) standard deviation across all 6 subjects in lateralization run

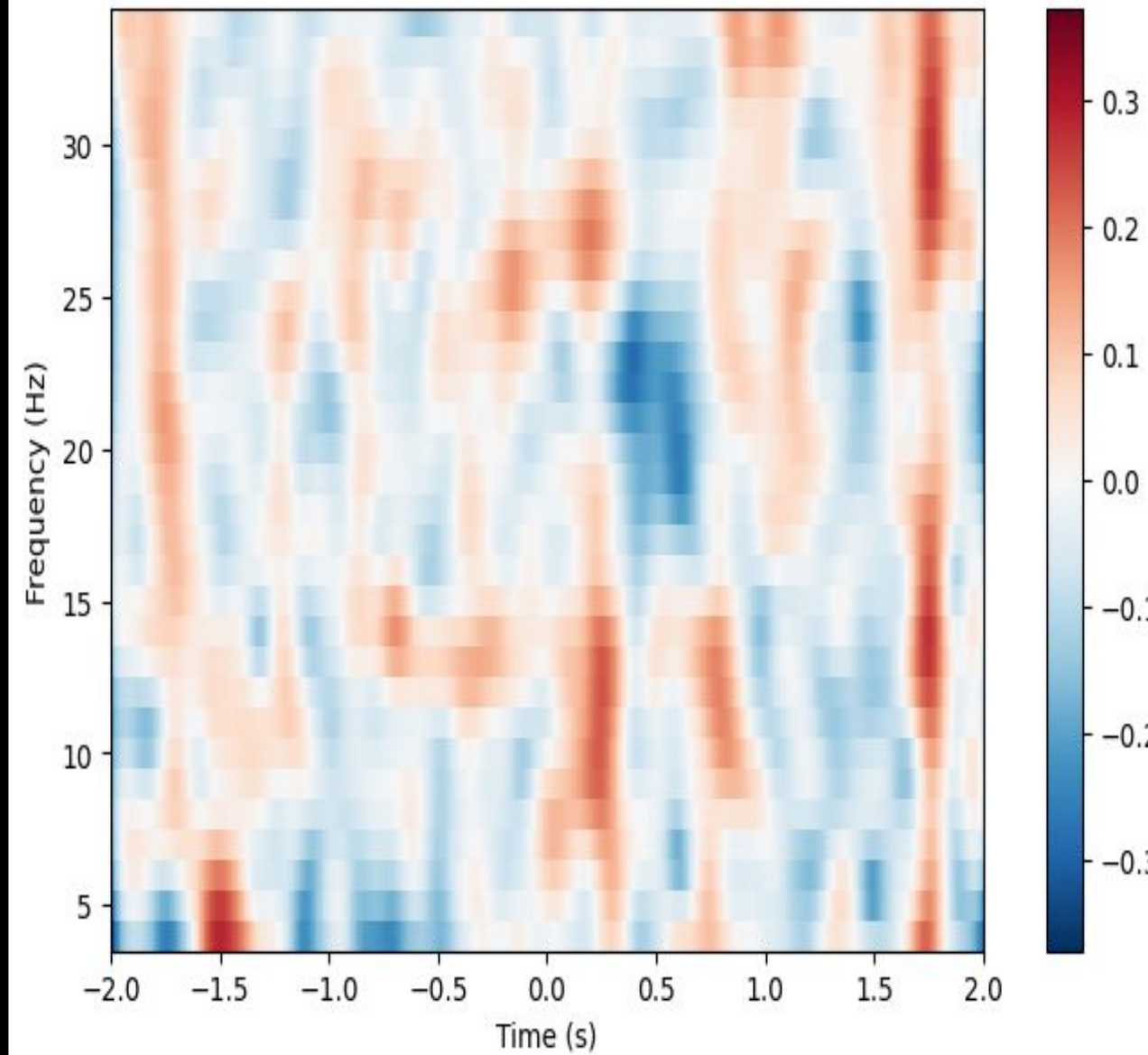


Time-Frequency Analysis

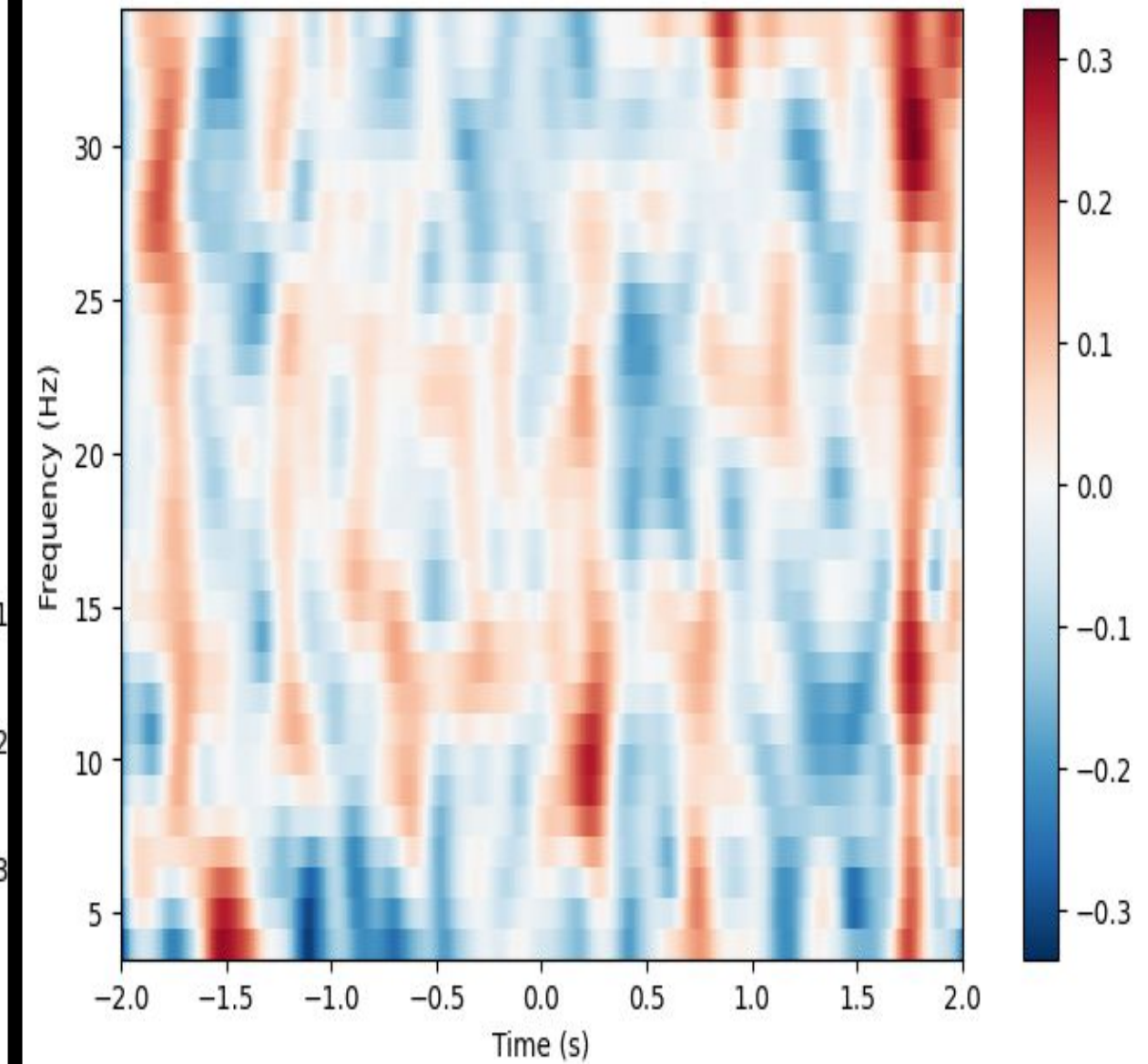
Time-Frequency (C3) Task (T1+T2)



Time-Frequency (Cz) Task (T1+T2)



Time-Frequency (C4) Task (T1+T2)



- the Red at the edges indicates Edge artifacts from morlet
- Time window [-2,2] seconds, baseline window = [-2,-1
- Time (0) seconds = task queued for subject to be still or imagine moving L/R hand, time [1,2] = beta rebound
-



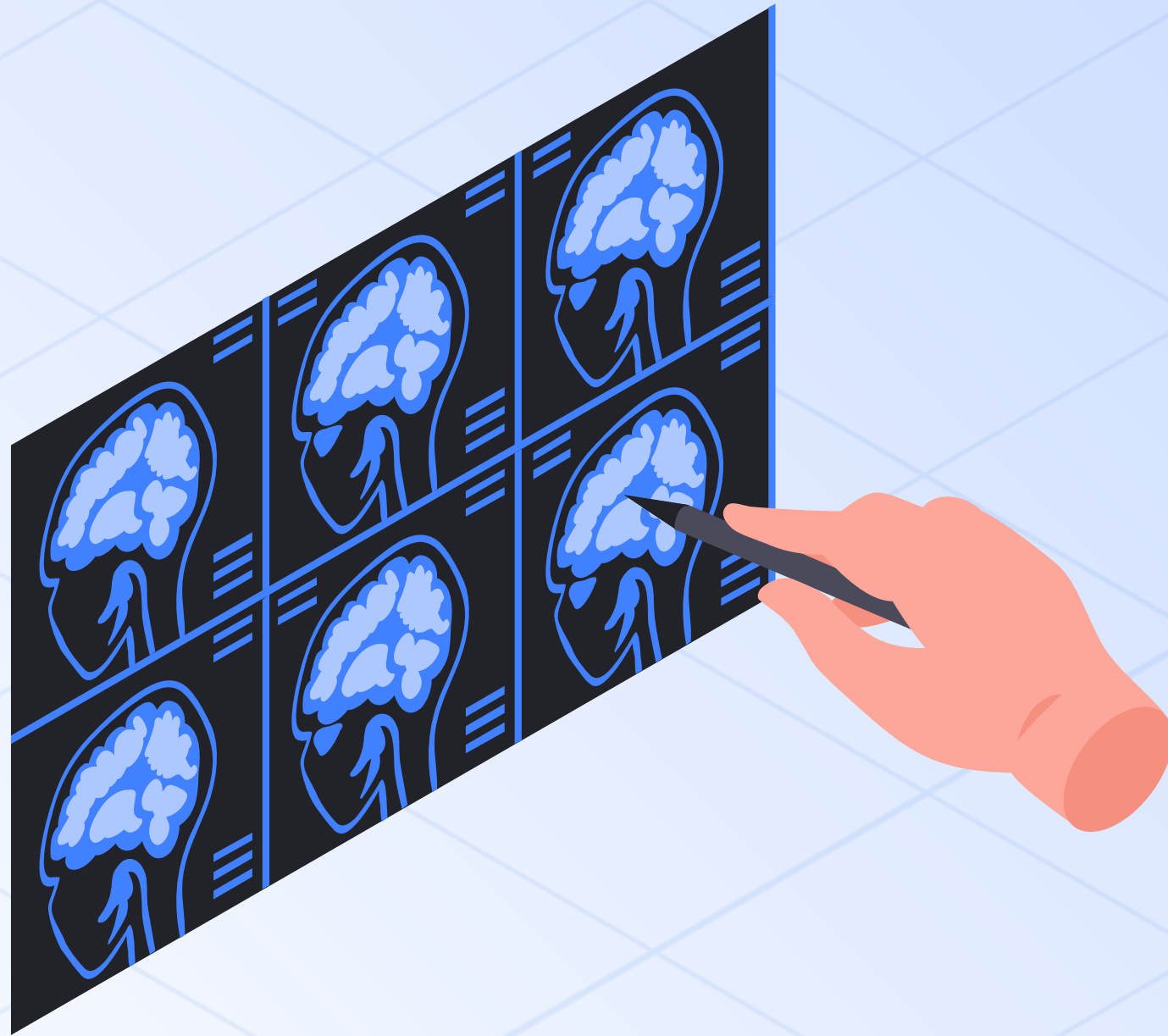
Conclusion

statistical results

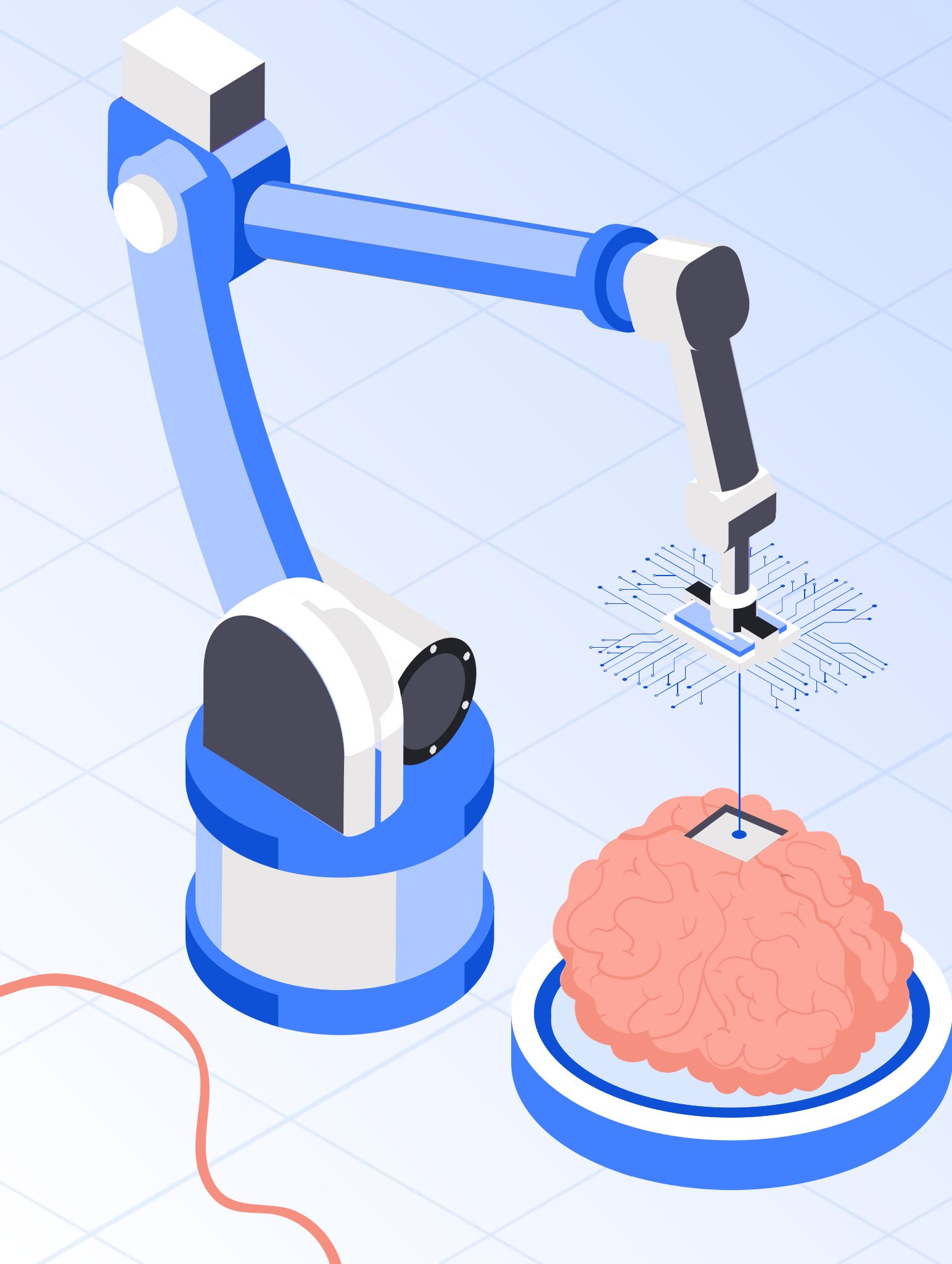


- Mu band
 - ERD $\approx 1.54\%$
 - $t \approx -0.13$, $p \approx 0.89$
 - VERY high variability
- Beta band
 - ERD $\approx 10.3\%$
 - $t \approx -1.71$, $p \approx 0.08$
- ERD calculated by comparing bandpower in task to relative baseline
- Used paired t-tests to compare baseline vs task bandpower across epochs
- Pattern consistent with motor-related ERD even if not typically significant

interpretations



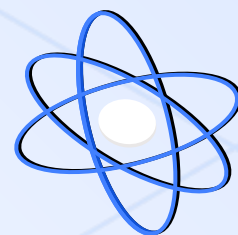
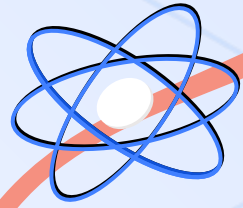
- comparing average bandpower differences
 - more variability in mu
 - beta is more reliable in this data
- decrease in mu and beta power suggest signal processing successful
- focus on electrodes C3, Cz, and C4 (primary motor cortex) in 10-20 EEG systems
 - provide strongest indications of motor activity



limitations

- artifacts can affect EEG signals
 - eye movements
 - muscle activity
 - breathing
 - environmental noise
- nearby electrode influence
 - electrical signals spread across scalp
- limited electrode selection
 - C1, C2

2026



BIG PICTURE

- BCIs
 - translating motor intent into device control
- Assistive Technology
 - prosthetic limbs
 - communication for paralyzed patients
- Neuroscience
 - understanding how brain plans movement

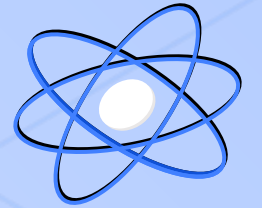
Future Work



- analyze more subjects and runs
- include surrounding electrodes
- apply bipolar subtraction
 - remove global noise from recordings
 - cleaner signal that shows localized activity
- use common spatial filtering (CSP)

Resources

Reference paper/Dataset Link : <https://physionet.org/content/eegmmidb/>



- **Goldberger, A., et al. "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation [Online]. 101 (23), pp. e215–e220." (2000). RRID:SCR_007345**
- Schalk, G., McFarland, D.J., Hinterberger, T., Birbaumer, N., Wolpaw, J.R. BCI2000: A General-Purpose Brain-Computer Interface (BCI) System. IEEE Transactions on Biomedical Engineering 51(6):1034-1043, 2004.
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- Pfurtscheller G, Brunner C, Schlögl A, Lopes da Silva FH. Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks. Neuroimage. 2006 May 15;31(1):153-9. doi: 10.1016/j.neuroimage.2005.12.003. Epub 2006 Jan 27. PMID: 16443377.

GitHub:

<https://github.com/keeratkaurt/sensorimotor-erd-eeeg-analysis>

Thank You!

